

A Note on Using Random Starting Values in Small Sample SEM

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Abstract

Model estimation for SEM analyses in commonly used software typically involves iterative optimization procedures, which can lead to nonconvergence issues. In this paper, we propose using random starting values as an alternative to the current default strategies. By drawing from uniform distributions within data-driven lower and upper bounds (see De Jonckere & Rosseel 2022), random starting values are generated for each (free) parameter in the model. Through three small simulation studies, we demonstrate that incorporating such bounded random starting values significantly reduces the nonconvergence rate, resulting in convergence rates ranging between 95% and 100% in the first two studies. In essence, bounded random starting values seem to offer a promising alternative to the default starting values that are currently used in most software packages.

A Note on Using Random Starting Values in Small Sample SEM

Introduction

Structural Equation Modeling (SEM) is a statistical method that allows to test (complex) relationships among latent variables. A prominent challenge in SEM is nonconvergence, which occurs when the optimizer fails to reach a stable solution. This issue is particularly prevalent when dealing with small sample sizes. Over the years, several solutions that try to tackle the problem of nonconvergence have been proposed. For instance, recent developments show that imposing simple bounds on the model parameters during estimation drastically decreases the amount of nonconverged cases and this without negatively affecting the point estimates of the unbounded parameters (De Jonckere & Rosseel, 2022; Lüdtke, Ulitzsch & Robitzsch, 2021). The simple lower and upper bounds proposed by the authors are (by default) derived directly from the data and are typically calculated for the (residual) variances of the observed variables, the (residual) variances of the latent variables, and the factor loadings.

An alternative to address the issue of nonconvergence is to use Bayesian estimation. Within the Bayesian framework, a crucial step involves specifying prior distributions for the model parameters. However, the process of selecting appropriate priors is often intricate. In fact, using default or diffuse priors can sometimes lead to worse performance compared to standard maximum likelihood estimation (Smid & Rosseel, 2020; Smid, 2023). Therefore, it is often recommended to perform a sensitivity analysis to assess the impact of those priors on the results (Van Erp, Mulder, Oberski, 2018).

Somewhat related to Bayesian estimation is shrinkage estimation of the sample variance-covariance matrix. In shrinkage estimation, a weighted average of the sample variance-covariance matrix and a highly structured shrinkage target is calculated. De Jonckere & Rosseel (2023) constructed a rationale for a model-based shrinkage target specifically designed for SEM analyses. This approach also allows researchers to incorporate their prior knowledge about the data, which can be translated directly into the model-based target matrix (De Jonckere & Rosseel, 2023). As with Bayesian

estimation, the accuracy of this prior knowledge is crucial for the performance of the model-based shrinkage target. In their study, De Jonckere & Rosseel (2023) demonstrated that the use of the model-based shrinkage target can effectively reduce the number of nonconverged cases compared to standard maximum likelihood estimation.

Nonconvergence cannot be solely attributed to a small sample size, as datasets with equivalent sample sizes may yield divergent results, with one dataset converging while another fails to do so. A crucial aspect of iterative optimization algorithms for model estimation is the determination of suitable (or optimal) starting values for the model parameters. Various strategies can be employed to determine these starting values. One approach involves adopting a "simple" approach whereby constants are selected as starting values. For example, starting values for factor loadings and (residual) variances can be set to 1, while all other parameters can be set to 0. Alternatively, other ad hoc values can be assigned such as assigning a value of 0.05 for latent variances or half the observed variance for residual variances of the indicators. Another approach involves utilizing non-iterative estimators to estimate (a subset of) the model parameters, and use these estimates as starting values. For the measurement part of the model, various non-iterative estimation procedures exist such as the Multiple Group Method (Guttman, 1952), Hägglund's instrumental variable approaches, i.e. FABIN1, FABIN2, and FABIN3 (Hägglund, 1982)¹, the non-iterative estimation procedure of Bentler (1982), among others (see Dhaene & Rosseel (2023) for an evaluation of non-iterative estimators in CFA). A final noteworthy approach is the use of *random* starting values, where values are somehow randomly selected for each free parameter in the model. The idea of using random starting values is a common one, and this strategy is used in many contexts. For example, the method is often applied in mixture (SEM) models to mitigate the risk of converging to a local maximum

¹ In lavaan 0.6-15, a combination of the "simple" approach and the approach using non-iterative estimators is applied per default. The factor loadings are initially estimated using FABIN3 (Hägglund, 1982), the observed residual variances are set to half the observed variance ($\frac{1}{2}s_{ii}^2$) and the latent variances are set to 0.05. Covariances and regression coefficients are initialized to zero.

or minimum (Shireman, Steinley, Brusco, 2015; Biernacki, Celeux, Govaert, 2003). On the other hand, Gilli & Schumann (2010) investigated the best-of-N-restarts strategy with random starting values in the field of finance and economics. A comparison between the best-of-N-restarts strategy (where N was set to 100) and a more "thought through" approach using constants that reflect reasonable values for the financial interpretation of the parameters in the model (see De Manousopoulos & Michalopoulos 2009, p 598), revealed that the utilization of random starting points yielded outcomes that were at least equivalent to, if not superior to, the strategy using constants.

Despite its prevalence in various contexts, random starting values are seldom employed in regular (non-mixture) SEMs. While generating random starting values may appear straightforward, generating 'good' random starting values can be challenging. Careful attention must be given to the selection of these values, including the choice of distribution from which they are drawn. Additionally, random starting values should be chosen to preserve the positive definiteness of the implied covariance matrix of the model. In this study, we propose choosing random starting values from a uniform distribution with a lower and upper bound. Defining the lower and upper bounds for the uniform distributions will be based on the work of De Jonckere & Rosseel (2022).

To the best of our knowledge, a review of the existing literature did not yield any papers that specifically investigate the process of defining starting values for regular (non-mixture) SEMs. Therefore, in this paper, we aim to investigate the effect of selecting various sets of random starting values on the convergence status of a model. The rest of the paper is organized as follows: subsequent sections of this paper will explain the concepts of model estimation in structural equation modeling and optimization algorithms. Furthermore, we will introduce the application of bounds to define initial parameter estimates and explore their effect on the convergence status of different models in three small simulation studies.

Model estimation in structural equation modeling

Assuming multivariate normal and complete data, $\mathbf{Y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the sample variance-covariance matrix, $\mathbf{S}$, and the mean vector $\bar{\mathbf{y}}$ are sufficient statistics. Because in this manuscript we assume centered data (i.e., $\boldsymbol{\mu} = \bar{\mathbf{y}} = \mathbf{0}$), $\mathbf{S}$ is the only summary statistic we need. A discrepancy function $F(\mathbf{S}, \boldsymbol{\Sigma}(\boldsymbol{\theta}))$ expresses the difference between the sample variance-covariance matrix and the model-implied variance-covariance matrix $\boldsymbol{\Sigma}(\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ denotes the free parameters in the model. Model estimation aims at finding those values for $\boldsymbol{\theta}$ that minimize the distance between $\boldsymbol{\Sigma}(\boldsymbol{\theta})$ and $\mathbf{S}$. Different discrepancy functions based on the least squares principle exist, but the maximum likelihood discrepancy function is probably the most commonly used method in SEM. This discrepancy function can be written as:

$$F_{ML}(\boldsymbol{\theta}) = \log|\boldsymbol{\Sigma}(\boldsymbol{\theta})| - \log|\mathbf{S}| + \text{tr}(\mathbf{S}\boldsymbol{\Sigma}(\boldsymbol{\theta})^{-1}) - P, \quad (1)$$

where $|\cdot|$ denotes the determinant of a matrix, $\text{tr}(\cdot)$ the trace of a matrix (i.e. sum of its diagonal elements), and P the number of observed variables. Model estimation in SEM as it is implemented in most statistical software is performed using an iterative optimization algorithm where an initial solution is improved through a series of iterations. The goal of the optimization algorithm is to find those values for $\boldsymbol{\theta}$ that minimize the discrepancy function in (1).

Optimization

In general, an optimization problem is defined as

$$\underset{\boldsymbol{\theta}}{\text{minimize}} F(\boldsymbol{\theta}), \quad (2)$$

where F is the objective function and $\boldsymbol{\theta}$ the parameter vector, subject to $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ where $\boldsymbol{\Theta}$ represents the parameter space². Different optimization algorithms apply different techniques to explore the parameter space and to find the best possible set of $\boldsymbol{\theta}$ values.

² Note that optimization could also include maximizing an objective function, by simply switching the sign of the objective function.

Roughly speaking, these optimization algorithms can be divided in two categories: gradient-based algorithms and gradient-free algorithms.

Gradient-based algorithms

Gradient-based or derivative-based algorithms utilize derivatives of the objective function while searching for the optimal solution. These algorithms differ in their utilization of either only the first derivative (gradient, $g(\boldsymbol{\theta})$), or both the first and second derivative (Hessian, $H(\boldsymbol{\theta})$). Gradient-based algorithms begin by selecting an initial point and evaluating the objective function at that specific point. The derivative is then computed to determine the new search direction. In the next step, the distance along the determined direction for the selection of the next point needs to be defined. This distance parameter is commonly known as the step length or the learning rate (α). The next point is then determined as a function of the current point, the search direction, and the step size. Iteratively, these initial two steps are repeated until certain convergence criteria are satisfied. Line search offers a means of selecting an optimal step size. Different types of line search exist: golden section search, bisection method, quadratic fit search, etcetera. Gradient-based methods in optimization typically employ four primary convergence checks or termination conditions (Kochenderfer & Wheeler, 2019). The first condition involves terminating the process if the maximum allowable number of iterations is exceeded. The second condition evaluates whether the change in the function value for successive iterations is smaller than a predetermined threshold, i.e.:

$$f(\boldsymbol{\theta}^{(k)}) - f(\boldsymbol{\theta}^{(k+1)}) < \epsilon. \quad (3)$$

If the change is indeed smaller than a specified threshold (ϵ), the process is terminated.

The third check also incorporates the change in the function value as in (3) but additionally utilizes the step factor in relation to the current function value, i.e.:

$$f(\boldsymbol{\theta}^{(k)}) - f(\boldsymbol{\theta}^{(k+1)}) < \epsilon |f(\boldsymbol{\theta}^{(k)})|. \quad (4)$$

The last check examines whether $g(\boldsymbol{\theta}^{(k+1)}) < \epsilon$. If all gradient values are lower than a certain threshold, the process is terminated. A zero gradient vector signifies the presence of a (local) minimum. Examples of first-order

methods (methods that use gradient information only) are gradient-descent and conjugate gradient (Kochenderfer & Wheeler, 2019). The Hessian matrix provides information about the curvature of a function at a specific point. This information plays a crucial role in determining the step length for a specific parameter. By understanding the magnitude of a small or large step in relation to a specific parameter (by knowing its scale), we can conduct a more effective search for the minimum. Second-order methods differ in whether the Hessian itself (Newton's method) or an approximation of the (inverse) Hessian is used (Quasi-Newton method). Calculating the Hessian at each iteration can be computationally intense, therefore, "Quasi-Newton" methods (for example, BFGS, Kochenderfer & Wheeler 2019) are often preferred in practice.

Gradient-free algorithms

Gradient-free algorithms provide an alternative approach to optimization that does not rely on derivatives for determining the search direction and step size. This characteristic makes them suitable for optimizing discrete or discontinuous problems where derivative-based methods may not be applicable. Various gradient-free algorithms have been developed, including genetic algorithms, simulated annealing, and the Nelder-Mead downhill Simplex method, among others (Kochenderfer & Wheeler, 2019). These algorithms differ in their strategies for bracketing, expanding, or shrinking the search region in pursuit of an optimal solution. By avoiding the need for derivative computations, gradient-free algorithms offer flexibility in tackling a wide range of optimization problems. Nevertheless, it is worth noting that these algorithms are generally perceived as being slower compared to their gradient-based alternatives.

Starting values and optimization

It is important to acknowledge that numerical optimization methods can be sensitive to the initial parameter values chosen at the start of the optimization process. This stands in contrast to closed-form solutions, which possess well-defined analytical expressions that can be derived algebraically and do not exhibit sensitivity to the initial

parameter values. How the starting values for the optimization process are chosen typically depends on the statistical software that is used.

Random starting values using bounds

When considering the use of random starting values, careful consideration must be given to the distribution from which these values are selected for the various parameters in the model. Establishing a distribution from which a random value for a parameter can be drawn is a nontrivial task, similar to the challenges faced in Bayesian estimation where priors for every parameter must be constructed. As mentioned in the introduction of this paper, De Jonckere & Rosseel (2022) proposed a bounded estimation approach. These bounds can be used as the lower and upper values for a uniform distribution, and have the potential to provide a suitable framework for choosing random starting values while preserving the positive definiteness of $\Sigma(\theta)$. In the bounded estimation approach, simple data-driven lower and upper bounds are defined for the model parameters. These simple bounds are constructed using the sample variances of the observed variables, hence the data-driven nature of the bounds. Starting from the one-factor model with p observed indicators, y_p , where $p = 1, 2, 3, \dots, P$ we have

$$y_p = \lambda_p f + \epsilon_p, \quad (4)$$

where λ_p denotes the factor loading for indicator p , f the latent factor, and ϵ_p the measurement error. Expressing both sides in terms of variances, we can write

$$\text{Var}(y_p) = \lambda_p^2 \text{Var}(f) + \text{Var}(\epsilon_p). \quad (5)$$

Because $\text{Var}(y_p)$ is observed, we can derive simple lower and upper bounds for the factor loadings, factor variances and the (residual) variances of the indicators. It is assumed that $\text{Var}(y_p)$ is strictly positive. Additionally, the bounds ensure that the sum of the elements on the right side of equation 5 is always nonnegative. Contingent upon the method that is used to set the metric of the latent variable(s), the lower and upper bounds for the factor loadings and the factor variance differ slightly. Table 1 and Table 2 summarize how these lower and upper bounds are defined when the variance of

the factor is fixed to unity or when the first factor loading is fixed to unity, respectively. It is worth noting that when employing the unit loading identification, the lower bound for the factor variance(s) cannot be set to zero. This constraint is necessary to ensure that finite bounds can be placed on the factor loadings within the model. Therefore, the lower bound for the factor variance is a function of a lower bound for a reliability of the marker indicator (see Table 2). Notably, De Jonckere & Rosseel (2022) did not put any bounds on the regression parameter(s) in the structural part. In this paper’s simulation studies, we will investigate the impact of using a zero starting value for β_{YX} .

The effect of random starting values using bounds

To evaluate the effect of choosing different starting values for nonconverged models, we generate data from a population model as shown in Figure 1, Figure 2 and Figure 5³. The population values are intentionally chosen to generate a high nonconvergence rate (e.g.: very low reliability of the indicators). In a first study, we look at the convergence rate and bias for a simple model. The second study evaluates these random starting values for a SEM model including crossloadings. The final simulation study looks at the effect of picking random starting values for a multistate-singletrait model (MSST model).

Study 1

For the initial simulation study, we generate one thousand random datasets, each with a sample size of 20, following the population model depicted in Figure 1. A structural equation model is fitted to each dataset. Using both the lavaan package (Rosseel, 2012) and the OpenMx package in R (Boker, S., Neale, M., Maes, H., et. al., 2011; Neale et al., 2016), only those cases that converge or fail to converge for both packages are extracted and used for further analysis⁴. A case is labeled ‘converged’ only when it satisfies two conditions: firstly, the convergence status is positive and secondly,

³ The code used for this paper is available for reference at <https://osf.io/8tguj/>.

⁴ By employing this approach, we aim to mitigate the influence of optimizer-specific characteristics that could potentially introduce variations in the convergence status.

all standard errors could be computed. The latter condition is an indirect check if $H(\theta)$ is positive definite. For cases where convergence is not achieved, we generate a maximum of 150 sets of random starting values for each case, and count the number of sets required to alter the convergence status of that particular nonconverged case.

For the model in Figure 1, we obtained a total of 360 nonconverged cases. The data presented in the top panel of Figure 3 indicates that out of the 360 nonconverged cases, the majority ($N = 272$) is able to achieve convergence after 15 or fewer random sets, where the starting value for β was set to zero. The top panel of Figure 4 indicates that already 119 cases converge for one random set of starting values and that almost all 360 cases converge within 150 random sets. These findings suggest that utilizing the bounded estimation approach to generate random sets of starting values is a highly efficient strategy, leading to an approximate 98% convergence rate for our initial model (Figure 1).

Attaining convergence is an important and desirable step, yet the question about the quality of the converged solution may arise. In order to assess the quality of the converged model, we examine the presence of improper solutions (e.g. negative variances) and the relative mean bias of $\hat{\beta}$. Results show that 243 out of 272 converged cases have at least one negative variance. The maximum number of observed negative variances for a specific case is two ($N=63$) and 29 cases have no negative variances. The mean bias for $N = 20$ and a population value for $\beta = 0.1$ is depicted in Table 3. Bias is compared between nonconverged cases where bounded estimation is applied, cases that converged in the first run (i.e. that did not need any random starting values) and cases where new starting values within bounds are constructed with a starting value for $\beta = 0$. The bias for the latter is based on cases that converged for 15 or fewer random starting value sets. As can be verified in Table 3, bias for cases where new starting values are constructed (with a starting value for $\beta = 0$) is lower than first run convergence. However, the value of zero is very close to the true value for β in the model, i.e. $\beta = 0.1$. Hence, we also considered a model where the true value equals 0.7 to evaluate the bias of a zero starting value for β . Surprisingly, bias for $\beta = 0.7$ and a

starting value for β equal to zero shows lower bias than for $\beta = 0.1$.

Study 2

In the second study, we consider a model containing several crossloadings (see Chen et al. 2001, p 477). As was the case for the first simulation study, we generate one thousand random datasets each with a sample size of 20, following the population model depicted in Figure 2. Again, a structural equation model is fitted to each dataset.

In the scenario involving a crossloadings model with a total of 341 nonconverged cases and a starting value for $\beta = 0$, the bottom panel of Figure 3 demonstrates that employing an alternative set of starting values enables successful convergence for the majority (70%) of these cases only for one random starting value set (86% converges within 30 or fewer random sets). Roughly around 6% of nonconverged cases for this specific model is not able to converge after 150 random sets of starting values. For the datasets that need at most 30 random starting value sets to converge, around 68% contain at least one negative variance. The maximum number of observed negative variances for a specific case is six ($N=2$).

Study 3

Since the initial model being evaluated in study 1 is rather simple, we further explore the effect of bounded random starting values for a more complex model: a MSST model (see Figure 5). The variance decomposition for the indicators in a MSST model is as follows:

$$\text{Var}(y_{it}) = \lambda_{T_{it}}^2 \text{Var}(\xi) + \lambda_{S_{it}}^2 \text{Var}(\zeta_t) + \text{Var}(\epsilon_{it}), \quad (6)$$

where $\text{Var}(\xi)$ the variance of the trait variable, $\lambda_{T_{it}}$ the trait loading, $\text{Var}(\zeta_t)$ the variance of the state variable, and $\lambda_{S_{it}}$ the state loading. Bounds are constructed in a similar way as before, where the latent state variables, ζ_t , are considered observed to come up with bounds for the trait factor loadings (i.e., $\lambda_{T_{it}}$). A thousand random datasets, each of sample size 10, are generated. After fitting a SEM to each dataset, we find 382 nonconverged cases. For those 382 cases, random starting value sets are

generated until convergence is reached (with a maximum allowed total of 150 sets).

Results show that about 64% of those 382 cases converge for a specific random starting value set. Only 16 of the converged cases show no improper solutions. The maximum number of observed negative variances for a specific case is four ($N=11$).

Discussion

Nonconvergence is a well-known problem in the field of small sample structural equation modeling. While the small sample size is often attributed as the cause, other underlying factors may also contribute. In SEM, model estimation involves an iterative procedure using optimization techniques, which necessitates the selection of appropriate starting values. The sensitivity of these techniques to the choice of starting values creates a possibility for nonconvergence when "poor" starting values are used. Although there exists a vast body of literature on optimization techniques and various approaches to determine starting values, the utilization of this knowledge in regular (non-mixture) SEMs remains limited. This paper aimed at bridging this gap by examining a specific method for determining starting values in regular SEMs: generating random starting values. These random starting values are drawn from a uniform distribution within specified lower and upper bounds as proposed by De Jonckere & Rosseel (2022).

Our simulation studies have demonstrated that generating random starting values within data-driven bounds effectively reduces the occurrence of nonconverged solutions. This approach led to a convergence rate of nearly 100% in both the first and second studies, indicating that for most of the datasets, at least one set of starting values could be identified that altered the convergence status. In the first study, for both lower and higher true values of β , the utilization of random starting values within the specified bounds (with the starting value for $\beta = 0$) resulted in reduced bias for $\hat{\beta}$ when compared to datasets that achieved convergence using default starting values in lavaan. A third study demonstrated that even for a more complex model, such as a multistate-singletrait model, the rate of nonconvergence decreased significantly, although to a lesser extent.

Overall, the implementation of bounded random starting values appears to offer a valuable alternative to the default starting values that are currently used in software. While the default starting values can only be employed once for a given dataset, bounded random starting values can be recalculated iteratively until convergence is achieved. It is noteworthy, however, that for the sake of computational efficiency, a reasonable threshold such as 50 different sets of starting values could be considered. Moreover, it is conceivable that several distinct random starting value sets could lead to convergence for a single dataset. Among these "successful" starting value sets, there might exist an optimal set that produces admissible solutions with minimal bias. However, determining such an optimal set appears to be considerably complex, necessitating further simulation studies. Future research could explore this intriguing aspect.

As a final remark, for certain datasets, bounded random starting values (and default starting values in lavaan) may fail to induce a change in the convergence status. This raises questions about the specific data characteristics contributing to this phenomenon. We are currently further exploring this.

Note that the issue of nonconvergence (and starting values) for SEM analysis disappears when non-iterative estimation procedures are employed. A recent simulation study underscores the potential efficacy of these techniques (Dhaene & Rosseel, 2023).

Open Practices Statement

All code (for analysis and data generation) is accessible through the Open Science Framework using the following link: <https://osf.io/8tguj/>.

Declarations

Funding

Not applicable

Conflicts of interest/Competing interests

None

Ethics approval

Not applicable

Consent to participate

Not applicable

Consent for publication

Not applicable

Availability of data and materials / Code availability

See Open Practices Statement above

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parameter	lower bound	upper bound
$\text{Var}(f)$	1	1
λ_p	$-\sqrt{\text{Var}(y_p)}$	$\sqrt{\text{Var}(y_p)}$
$\text{Var}(\epsilon_p)$	0	$\text{Var}(y_p)$

Table 1

Table extracted from De Jonckere & Rosseel (2022). Standard lower and upper bounds when fixing the variance of the factor to unity for identification.

parameter	lower bound	upper bound
$\text{Var}(f)$	$\psi^{(l)}$	$\text{Var}(y_1)$
λ_1	1	1
λ_p	$-\sqrt{\text{Var}(y_p)/\psi^{(l)}}$	$\sqrt{\text{Var}(y_p)/\psi^{(l)}}$
$\text{Var}(\epsilon_p)$	0	$\text{Var}(y_p)$

Table 2

Table extracted from De Jonckere & Rosseel (2022). Standard lower and upper bounds when fixing the first factor loading to unity for identification. The lower bound for the factor variance $\psi^{(l)}$ is defined as

$$\psi^{(l)} = \text{Var}(y_1) - [1 - \text{Rel}(y_1)] \text{Var}(y_1),$$

where $\text{Rel}(y_1)$ is a user-specified value for the reliability of the marker indicator y_1 .

β	ML_c	start_0	bounds
0.1	-0.51	-0.33	-0.06
0.7	-0.29	0.18	-0.59

Table 3

Bias for first run convergence with standard ML (ML_c), cases where new starting values within bounds are constructed with a starting value for $\beta = 0$ (start_0), and bias for model estimation using bounded estimation (bounds). The top panel contains values for a model where $\beta = 0.1$, whereas the bottom panel contains values for a model where $\beta = 0.7$.

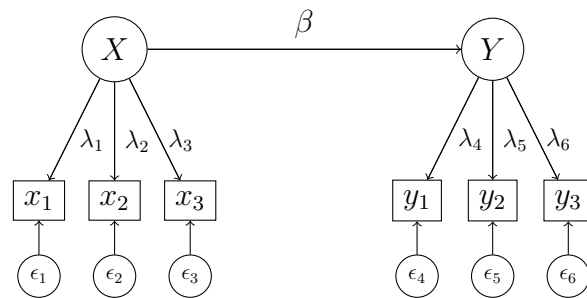


Figure 1. A simple model where the latent variable Y is regressed on the latent variable X . Each latent variable is measured by three observed indicators. The population values are as follows: $\lambda_1 = \lambda_4 = 1$, $\lambda_2 = 0.8$, $\lambda_3 = 0.6$, $\lambda_5 = 0.7$, $\lambda_6 = 0.9$. The latent (residual) variances are set to 1.0 and the value for β equals 0.1.

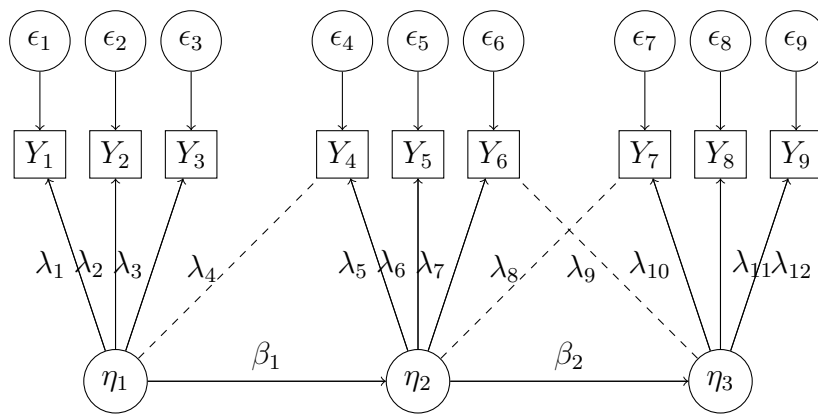


Figure 2. The population model used for study 2. The population values for β_1 and β_2 are 0.1 and 0.2, respectively. For the factor loadings, the population values are fixed at 1.0, except for the crossloadings ($\lambda_4, \lambda_8, \lambda_9$) which have a population value of 0.3. The latent (residual) variances are set to 1.0. The residual variances of the indicators are 0.51 for the y_1, y_2, y_3, y_5, y_8 and y_9 indicators, and 0.2895 for the y_4, y_6 and y_7 indicators.

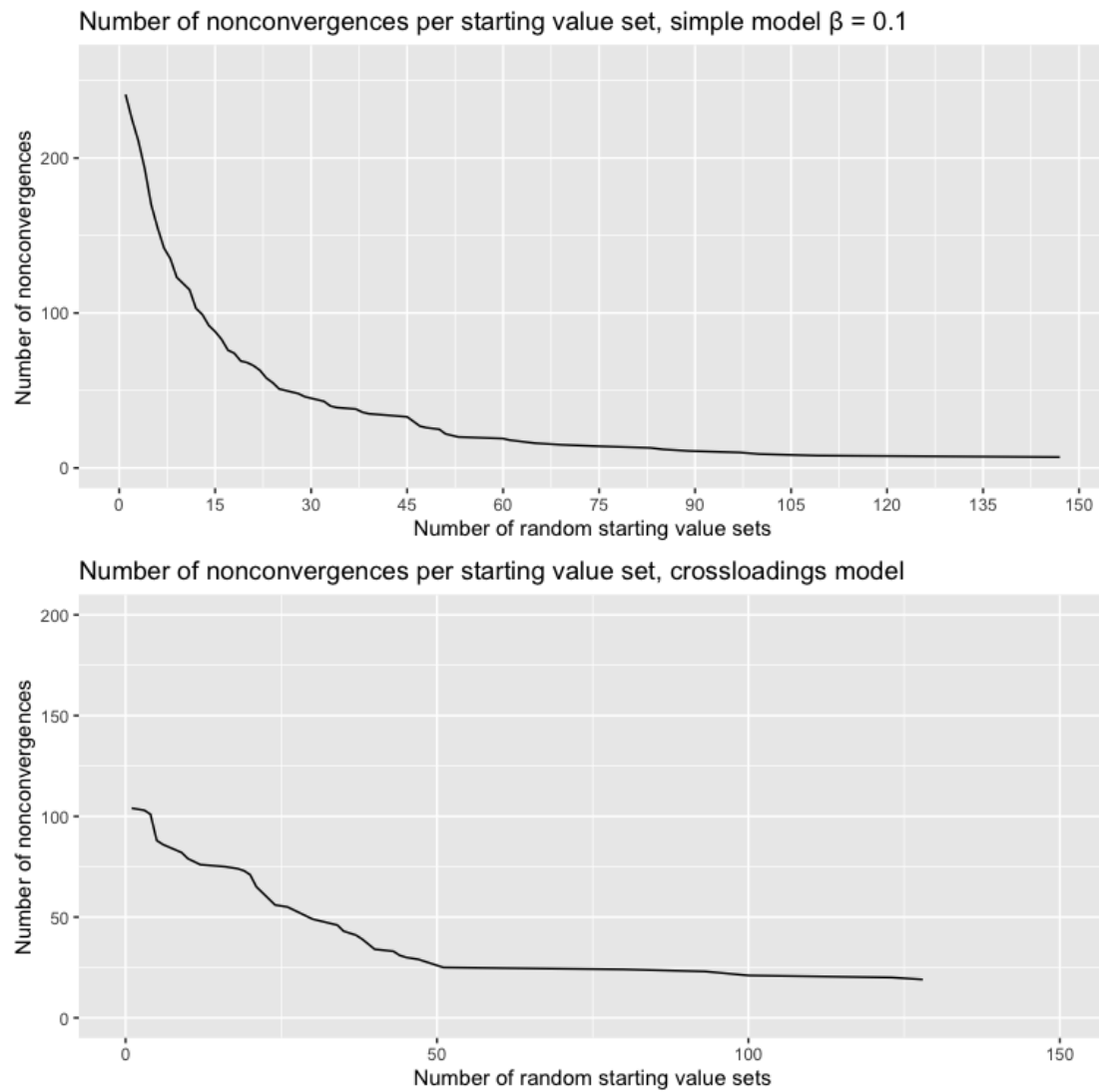


Figure 3. Number of nonconvergences per new random starting value set for a model as in Figure 1 (top panel) and as in Figure 2 (bottom panel). The total number of nonconvergences for the first model is 360 and 341 for the second model. On the Y-axis, the number of nonconverged cases for a given number of random sets (X-axis) is found.

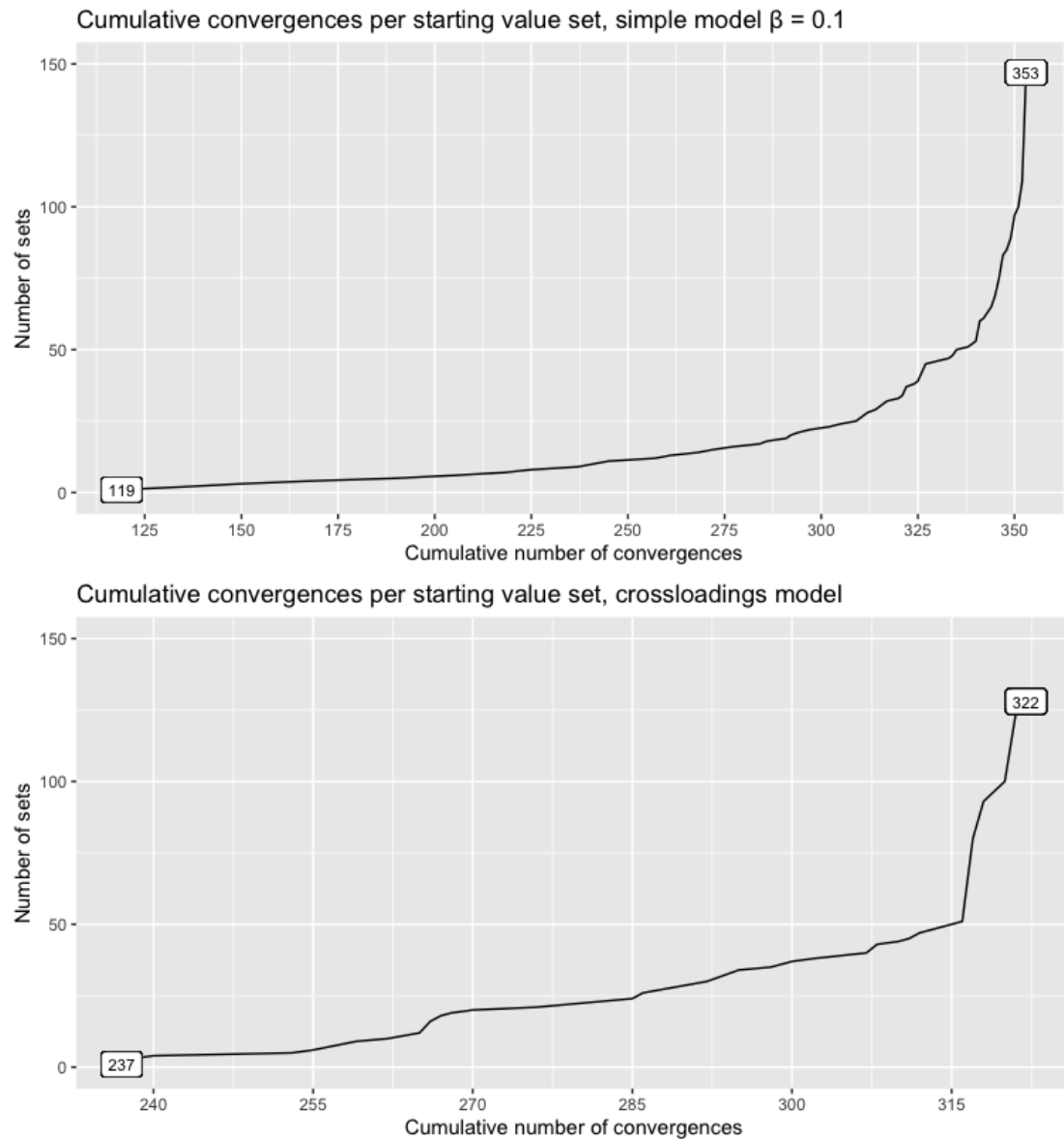


Figure 4. The cumulative number of converged cases for one to 150 random starting value sets. Both for a model as in Figure 1 (top panel) and as in Figure 2 (bottom panel). A total of 353 out of 360 cases converge for model one, whereas 322 out of 341 cases converge for new starting values set in model two.

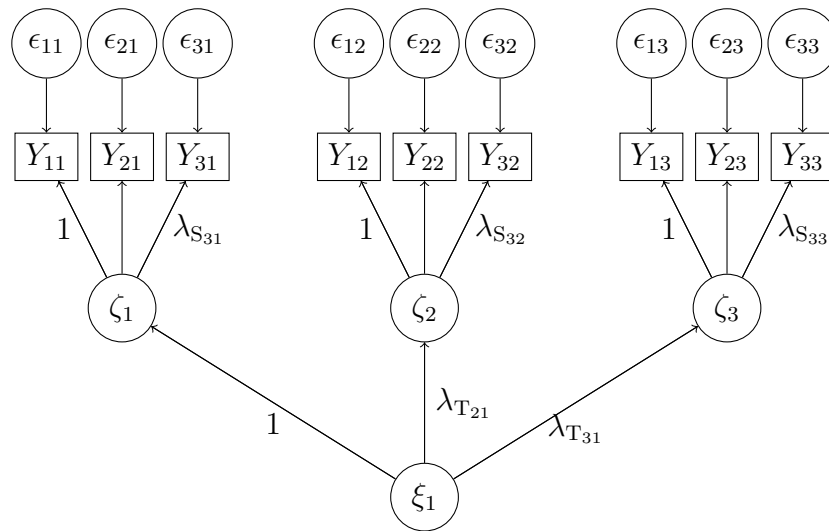


Figure 5. Simplified representation of a multitrait-singlestate (MSST) as a second-order CFA model (or a single-level SEM) for $t = 3$ timepoints, where ζ_i reflects the state variable and ξ_i the trait variable. The factor loadings are set to $\lambda_{S_{1t}} = \lambda_{T_1} = 1$, $\lambda_{S_{2t}} = \lambda_{T_2} = 0.5$, and $\lambda_{S_{3t}} = \lambda_{T_3} = 1.3$. The measurement errors are set to $\epsilon_{1t} = 0.6$, $\epsilon_{2t} = 0.25$, and $\epsilon_{3t} = 0.7$. The remaining true values can be consulted in the osf repository.